

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 8

Question 1 - MCQ · PG-53 · LR-1

The chi-square statistic is denoted by

- A. χ^2
- B. σ^2
- C. r
- D. μ

Correct answer: χ^2

Explanation and steps

The standard symbol is χ^2 .

Question 2 - MCQ · PG-53 · LR-2

The chi-square statistic is computed using

- A. $\Sigma((O - E)^2/E)$
- B. $\Sigma(O + E)$
- C. $\Sigma(O - E)$
- D. E/O only

Correct answer: $\Sigma((O - E)^2/E)$

Explanation and steps

This is the textbook formula.

Question 3 - MCQ · PG-53 · LR-3

Expected frequency is calculated as

- A. row total $\times$ column total / N
- B. row total + column total
- C. N / observed frequency
- D. observed $\times N$

Correct answer: row total $\times$ column total / N

Explanation and steps

Use $E_{ij} = R_i C_j / N$.

Question 4 - MCQ · PG-53 · LR-4

O in the chi-square formula denotes

- A. observed frequency
- B. expected frequency
- C. overall mean
- D. order number

Correct answer: observed frequency

Explanation and steps

O means observed.

Question 5 - MCQ · PG-53 · LR-5

E in the chi-square formula denotes

- A. expected frequency
- B. observed frequency
- C. extreme value
- D. error only

Correct answer: expected frequency

Explanation and steps

E means expected.

Question 6 - MCQ · PG-53 · LR-6

N in the expected-frequency formula denotes

- A. grand total
- B. row total
- C. column total
- D. number of classes only

Correct answer: grand total

Explanation and steps

N is the total frequency.

Question 7 - MCQ · PG-53 · LR-7

The value of χ^2 can never be

- A. negative
- B. zero
- C. positive
- D. a decimal

Correct answer: negative

Explanation and steps

Every term is non-negative.

Question 8 - MCQ · PG-53 · LR-8

If $O = E$ for every cell, χ^2 is

- A. 0
- B. 1
- C. -1
- D. N

Correct answer: 0

Explanation and steps

All squared differences are zero.